Note

(i) When no exponent is explicitly shown in a variable of a term, it is understood to be 1. For example, $11p = 11p^1$.

(ii) For an expression in x, if the terms of the expressions are in descending powers of x and the like terms are added, then we say that it is in the standard form.
For example, $x^4 - 3x^3 + 5x^2 - 7x + 9$ is in the standard form. It is easier to find the highest power term when the expression is in standard form. Highest power of this expression is 4.

(iii) The highest degree term of an algebraic expression is called as leading term.

Let us consider an algebraic expression: $x^3 - 3x^2 + 4$.

The terms of the above expression are x^3, $-3x^2$ and 4. Exponent of the term x^3 is 3 and $-3x^2$ is 2.

Thus, the term x^3 has the highest exponent, that is 3.

Now, consider the expression, $3x^4 - 4x^3y^2 + 8xy + 7$.

Take each term and check its power. In $3x^4$, exponent is 4, hence its degree is 4. In $-4x^3y^2$, the sum of powers of x and y is 5, hence its degree is 5. In $8xy$, the sum of powers is 2.

Therefore, the term with highest power in the above expression is $-4x^3y^2$ and its power is 5, which is called as degree of this expression.

The term(s) containing the highest power of the variables in an expression is called the degree of expression.

The degree of any term in an expression can only be a positive integer. Also, degree of expression doesn't depend on the number of terms, but on the power of variables in the individual terms. The degree of constant term is 0.

Try these

1. Complete the following table:

S. No.	Algebraic expression	Degree of the terms				Degree of the expression
		Term-I	Term-II	Term-III	Term-IV	
1.	$7x^3 - 11x^2 + 2x - 5$	3	2	1	0	3
2.	$9x^5 - 4x^2 + 2x - 11$	5	2	1	0	5
3.	$6b^2 - 3a^2 + 5a^2b^2$					
4.	$p^4+p^3+p^2+1$					
5.	$6x^2y^3 - 7x^3y + 5xy$					
6.	$9 + 2x^2 + 5xy - 5x^3$					

7th_Maths_T2_EM_Chp3.indd 59 09-07-2019 12:21:03